

4(a)	<i>any 4 of:</i> 1 spontaneous / random, mutation ; 2 produces an allele coding for enzyme (that gives resistance) ; 3 toxin is selection pressure ; 4 flies or larvae, with, allele / enzyme / mutation, have selective advantage ; 5 (flies with, allele / enzyme / mutation) survive / reproduce ; ora 6 pass on this allele (to offspring) ; 7 this allele increases in frequency ;	4
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Question	Answer	Marks
4(b)	<i>any 3 of:</i> 1 separated by a geographical barrier / description ; 2 different, environmental conditions / selection pressures ; 3 allopatric speciation ; 4 reproductive isolation / no breeding with mainland flies / no gene flow (between two populations) ; A pre-zygotic isolation 5 new allele not, lost / diluted in frequency ; 6 founder effect / genetic drift ; 7 (more pronounced) in small population ;	3
4(c)	<i>any 3 of:</i> 1 obtain, DNA / genome / gene / base / nucleotide, sequences (from database) ; 2 obtain, amino acid / protein, sequences (from database) ; 3 compare sequences (of two populations) ; 4 calculate percentage of, bases / amino acids, changed ; 5 mitochondrial DNA ; 6 molecular clock / mutation rate, (to estimate time) ;	3